

## Course Prerequisites

This course requires you to be proficient in data analysis and have experience using SQL, Relational Databases, performing data collection, data wrangling, data analysis, & data visualization with Python libraries, and using a BI tool like IBM Cognos Analytics or Google Looker. It is recommended that you complete all prior courses in the [IBM Data Analyst Professional Certificate](#) before starting this project course:

- Introduction to Data Analytics
- Excel Basics for Data Analysis
- Data Visualization and Dashboards with Excel and Cognos
- Python for Data Science, AI & Development
- Python Project for Data Science
- Databases and SQL for Data Science with Python
- Data Analysis with Python
- Data Visualization with Python

## Course Syllabus

### Module 1: Welcome to the Course

- Welcome
- Collecting Data Using APIs
- Collecting Data Using Web Scraping
- Exploring Data

### Module 2: Data Wrangling

- Assignment Overview
- Finding Duplicates
- Removing Duplicates
- Finding Missing Values
- Imputing Missing Values
- Normalizing Data
- Data Wrangling

### Module 3: Exploratory Data Analysis

- Assignment Overview
- Analyzing the Data Distribution

- Handling Outliers
- Correlation

#### **Module 4: Data Visualization**

- Assignment Overview
- Visualizing Distribution of Data
- Visualizing Relationship
- Visualizing Composition of Data
- Visualizing Comparison of Data

#### **Module 5: Building a Dashboard**

- Assignment Overview
- Dashboards

#### **Module 6: Final Assignment: Present Your Findings**

- How to Present Your Findings
- Final Presentation
- Course Wrap Up